Lab # 02

Task - 1: Program 1

```
1# include <iostream
2. Using namespace std;
3- int main ()
4. {
5- char letter
6- letter = 'A';
7- Cout << letter << '\n';
8- Letter = 'B';
9- Cout << letter << '\n';
10- cin >> letter;
11- return 0;
12- }
```

output: A
B

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Task-1: Program 2

```
1- # include <iostream>
2- Using namespace std;
3- int main ()
4- {
5- char letter;
6- letter = 65;
7- cout << letter << '\n';
8- letter = 66
9- cout << letter << '\n';
10- Cin >> letter;
11- return 0;
12- }
```

Output: A
B

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Task-1 Program 3

```
1-# include <iostream>
2- Using namespace std;
3- int main ()
4- {
5- int letter
6- Letter = 'A';
7- Cout << letter << '\n';
8- Letter = 'B';
9- Cout << letter << '\n';
10- Cin >> letter;
11- return 0;
12.}
```

Output:

65

66

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Task - 2:

```
1  #include <iostream>
2  Using namespace std;
3  int main ()
4  {
5  int a, b;
6  cout << "Enter two integers";
7  cin >> a >> b
8  cout << a << "+" << b << " = " << a+b
   << endl;
9  cout << a << " - " << b << " = " << a-b
   << endl;
10 cout << a << " * " << b << " = " << a * b <<
   endl;
11 cout << a << " / " << b << " = " << a/b <<
   endl;
12 return 0;
13 }
```

Output :

```
Enter two integers : 10 5
10 + 5 = 15
10 - 5 = 5
10 * 5 = 50
10 / 5 = 2
```

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Task-3:

```
1- # include stdafx h"
2- # include <iostream>
3- Using namespace std;
4- int main ()
5- {
6- int val1;
7- int val2;
8- int val3;
9- Cout << "please enter your 3 numbers";
10- Cin >> val1 >> val2 >> val3;
11- Cout << val1 "<< /n"<< val2 <<"/n << val3 <<
    "/n";
12- Cout << val3 << "\n"<< val2 <<"/n"<< val1 <<
    "/n";
13- System ("Pause");
14- return 0;
15- }
```

Output:

```
please enter your 3 number: 1,2,3
1
2
3
3
2
1
```

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Task - 4

```
1- # include <iostream>
2- # include <math>
3- Using namespace std;

4- int main () {
 5-   double x1, y1, x2, y2;
 6- Cout << " Enter the x and y coordinate of the first points;
 7- Cin >> x1 >> y1;
8- Cout << " Enter the x and y coordinate of the second points;

9- Cin >> x2 >> y2;
10- double distance = Sqrt(Pow(x2-x1, 2) + Pow(y2-y1, 2));
11- Cout << "distance between the points " << distance << "/n";

12- return 0;

13- }
```

Output:

Enter x and y coordinate of the first point 1 2
Enter the x and y coordinate of the second point 4 6
Distance between the points 5

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Task - 5

```
1-# Include <iostream>
2-Using namespace std;
3- int main () {
4- Constant double PI = 3.14159265358979793;
5- double radius , length, width, base , height;
6- Cout << "Enter the circle's radius";
7- Cin >> radius;
8-Cout << "Enter the rectangle's length and width
9- Cin >> length >> Width;
```

Output:

```
Enter the circle radius : 3
Enter the rectangle's length and width 5,4
Enter the triangle's base and height : 6 4
Circle area : 28.2743
Rectangle area : 20
Triangle area : 12
```

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Task-6

```
1- #include <iostream>
2- # include <iomanip>
3- Using namespace std;
4- int main ( ) {
5- double item 1 = 12.95;
6- double item 2 = 24.95;
7- double item 3 = 6.95;
8- double item 4 = 1495;
9- double item 5 = 3.95;
10- double Subtotal = item1+item2+item3+Item4+ item5;
11- double SalesTax = Subtotal * 0.06,
12- double Total1 = Sales Tax
13- Cout<<fixed << Setprecision (2);
14- Cout << "Item 1 price: $" << item 1 <<'/n;"
15- Cout << " Item 2 Price: $" << item 2<<'/n;"
16- Cout << "Item 3 Price: $ << item 3 <</n;
17- Cout << " Item 4 Price: $ << item 4 <<'/n,
18- Cout << " Item 5 Price : $ << item 5 <<" /n";
19- Cout << subtotal $" << subtotal <<'/n;
20- Coutt<Sale tax (6%) : $" << SalesTax<<'/n;
21- Cout<<" Total: $ "<<total <<'/n;
22- return 0;
23 }
```

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Output:

Item 1 Price : $ 12.95
Item 1 Price : $ 12.95
Item 2 price : $ 24.95
Item 3 Price : $ 6.95
Item 4 Price : $ 14.95
Item 5 Price : $ 3.95
Subtotal : $ 63.75
Sales tax (6%) = $ 3.83
Total : $ 67.58

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Bonus-Task

```
1- # include <iostream>
2- Using namespace std;
3- int main ( ) {
4-    int number;
5- Cout <<"Enter a five digit integer: ";
6- Cin >> number;
7- int digit 1 = number / 10000;
8- int digit 2 = (number / 1000) % 10;
9- int digit 3 = (number / 100) % 10;
10- int digit 4 = (number / 10) % 10;
11- int digit 5 = number % 10;
12- Cout << digit 1 <<"   "
13- << digit 2 <<"   "
14- << digit 3 <<"   "
15- << digit 4 <<"   "
16- << digit 5 <<'\n';
17- return 0
18- }
```

Output

```
Enter a five digit integers: 12345

1   2   3   4   5
```

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Home Task 1:

```
1- # include <iostream>
2- Using namespace std;
3- int main ( )
4- {
   5- int var1 , var2 tem;
   6- Cout <<"Enter the first
7- value: ";
    8- Cin >> var 1;
    9- Cout << "Enter the second
    10-Values:";
     11- Cin >> var 2;
    12- Cout << " Before swapping:"
    13- << end 1;
    14- Cout << "var1 = "<< var1 <<
    15- end 1;
     16- Cout << "Var2 = "<<var2 <<
     17- end 1;
      18- temp = var1;
      19- var 1 = var 2;
       20- Var 2 = temp;
       21- Cout << "After swapping:"<<
     22end1;
        23- cout << " var 1 = "<< var 1 <<
```

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```
end 1;
24- cout << "var2 = " << var2 <<
25- end 1;
26- return 0;
27- }
```

Output:

```
Enter the first value: 10
Enter the Second value: 20
Before swapping:
var1 = 10
var2 = 20
After Swapping:
var1 = 20
var2 = 10
```

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Home Task 2

```
1- # include <iostream>
2- Using namespace std;
3- int main ()
4- {
5-    char letter;
6-  cout << "Enter a lowercase
7- alphabet: ";
8-    cin >> letter;
9-    char uppercase = letter
10-  32;
11-   Cout << "uppercase alphabet:
12- " << uppercase << end 1;
13- return 0;
14- }
```

Output:

Enter a lowercase alphabet : a

uppercase Alphabet : A

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Task 01

main.cpp

Run

```cpp
#include <iostream>
using namespace std;

int main() {
    char letter;

    letter = 'A';
    cout << letter << '\n';

    letter = 'B';
    cout << letter << '\n';

    cout << "Enter a character: ";
    cin >> letter;
    cout << "You entered: " << letter << '\n';

    return 0;
}
```

Output

```
A
B
Enter a character:
```

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Task 02

```cpp
#include <iostream>
using namespace std;

int main() {
    int letter;

    letter = 'A';
    cout << letter << '\n';

    letter = 'B';
    cout << letter << '\n';

    cout << "Enter a character: ";
    cin >> letter;
    cout << "You entered: " << letter << '\n';

    return 0;
}
```

```
65
66
Enter a character:
```

Task 03

main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;

int main() {
    int val1, val2, val3;

    cout << "Please enter your 3 numbers: ";
    cin >> val1 >> val2 >> val3;

    cout << val1 << '\n' << val2 << '\n' << val3 << '\n';
    cout << val3 << '\n' << val2 << '\n' << val1 << '\n';

    return 0;
}
```

Output

```
Please enter your 3 numbers:
```

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Task 04

main.cpp — Run

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    double x1, y1, x2, y2;

    cout << "Enter the x and y coordinates of the first point: ";
    cin >> x1 >> y1;

    cout << "Enter the x and y coordinates of the second point: ";
    cin >> x2 >> y2;

    double distance = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

    cout << "Distance between the points: " << distance << '\n';

    return 0;
}
```

Output

```
Enter the x and y coordinates of the first point:
```

Task 05

main.cpp — Run

```cpp
#include <iostream>
using namespace std;

int main() {
    const double PI = 3.141592653589793;

    double radius, length, width, base, height;

    cout << "Enter the circle's radius: ";
    cin >> radius;

    cout << "Enter the rectangle's length and width: ";
    cin >> length >> width;

    cout << "Enter the triangle's base and height: ";
    cin >> base >> height;

    double circleArea = PI * radius * radius;
    double rectangleArea = length * width;
    double triangleArea = base * height * 0.5;

    cout << "Circle area: " << circleArea << '\n';
    cout << "Rectangle area: " << rectangleArea << '\n';
    cout << "Triangle area: " << triangleArea << '\n';

    return 0;
}
```

Output

```
Enter the circle's radius:
```

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Task 06

main.cpp

Run

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal = item1 + item2 + item3 + item4 + item5;
    double salesTax = subtotal * 0.06;
    double total = subtotal + salesTax;

    cout << fixed << setprecision(2);

    cout << "Item 1 price: $" << item1 << '\n';
    cout << "Item 2 price: $" << item2 << '\n';
    cout << "Item 3 price: $" << item3 << '\n';
    cout << "Item 4 price: $" << item4 << '\n';
    cout << "Item 5 price: $" << item5 << '\n';
    cout << "Subtotal: $" << subtotal << '\n';
    cout << "Sales tax (6%): $" << salesTax << '\n';
    cout << "Total: $" << total << '\n';

    return 0;
}
```

Output

```
Item 1 price: $12.95
Item 2 price: $24.95
Item 3 price: $6.95
Item 4 price: $14.95
Item 5 price: $3.95
Subtotal: $63.75
Sales tax (6%): $3.82
Total: $67.58
```

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Bonus Task

main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;

int main() {
    int number;

    cout << "Enter a five-digit integer: ";
    cin >> number;

    int digit1 = number / 10000;
    int digit2 = (number / 1000) % 10;
    int digit3 = (number / 100) % 10;
    int digit4 = (number / 10) % 10;
    int digit5 = number % 10;

    cout << digit1 << "   "
         << digit2 << "   "
         << digit3 << "   "
         << digit4 << "   "
         << digit5 << '\n';

    return 0;
}
```

Output

```
Enter a five-digit integer:
```

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Home Task 01

main.cpp

Run

```cpp
#include <iostream>
using namespace std;

int main()
{
    int var1, var2, temp;

    cout << "Enter the first value: ";
    cin >> var1;

    cout << "Enter the second value: ";
    cin >> var2;

    cout << "Before swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    return 0;
}
```

Output

```
Enter the first value: 10
Enter the second value: 20

Before swapping:
var1 = 10
var2 = 20

After swapping:
var1 = 20
var2 = 10
```

Home Task 02

main.cpp

Run

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter;

    cout << "Enter a lowercase alphabet: ";
    cin >> letter;

    char uppercase = letter - 32;

    cout << "Uppercase alphabet: " << uppercase << endl;

    return 0;
}
```

Output

```
Enter a lowercase alphabet:  a
Uppercase alphabet: A
```

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